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Paleomagnetism and the Late Jurassic Genesis of the Illinois-Kentucky Fluorspar Deposits

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Abstract

The Illinois-Kentucky fluorspar district has yielded most of the United States fluorite production. The fluorspar ore occurs as epigenetic replacement deposits in Mississippian limestone and as vein deposits. Paleomagnetic analysis was done on 324 specimens from 33 sites in fluorspar ore and in host limestones and cap-rock sandstones both adjacent to and remote from ore using alternating field and thermal step demagnetization. Saturation isothermal remanence methods were used to identify magnetization carriers in typical specimens and mineral crystals. An *A* characteristic remanent magnetization (RM) component was found outside the district in stratigraphically equivalent limestones that record a Pennsylvanian Kiaman remagnetization, and a *C* characteristic RM was found in surface weathered specimens that records the Recent earth's magnetic field. At 28 sites in ore and nearby host rocks a *B* characteristic RM was found that gives a Late Jurassic pole position of 152.4° E, 78.0° N (dp = 3.0°, dm = 4.4°). Late Jurassic genesis for the fluorspar fits within the permissible geologic window from Early Permian to Late Cretaceous and agrees with fission track and strontium isotope dating on the fluorites but does not agree with a recent Sm/Nd date. It is suggested that a Late Jurassic hydrothermal event, here termed the "Rosiclare event," formed the fluorspar deposits, probably remagnetized Kiaman-remagnetized host rocks, probably was controlled geographically by faults of the Reelfoot-Wabash rift system, and probably coincided with the substantial uplift in the region that formed the Pascola arch.

Introduction

ABOUT 75 percent of the fluorspar that has been mined in the United States has come from vein and replacement deposits in Mississippian host rocks of the Illinois-Kentucky fluorspar district (Baxter et al., 1989; Fig. 1). Dating of these ores has been hampered by the lack of suitable minerals for conventional radiometric methods. The genesis of these deposits has been related to different events such as the emplacement of nearby Permian alkalic intrusions (e.g., Zartman et al., 1967) or to the mixing of subsurface fluids of uncertain age (e.g., Richardson and Pinckney, 1984). The object of this study was to use paleomagnetic methods to date the ore and, thereby, to constrain geologic models for genesis of the fluorspar ore.

Geology

The Illinois-Kentucky fluorspar district covers an area of about 2,500 km^2 straddling the southern Illinois-western Kentucky boundary (Fig. 1). Baxter et al. (1989) have recently summarized the extensive geologic literature of the district.

The Precambrian basement of the Illinois-Kentucky fluorspar district is overlain by Cambrian to Pennsylvanian sedimentary rocks that host two end-member types of epigenetic fluorspar deposits—manto or bedding replacement deposits that grade into fissure-filling vein deposits (Grogan and Bradbury, 1968). The manto deposits occur mostly in three middle Mississippian horizons, namely, the sub-Rosiclare ore in limestone beneath the Spar Mountain Sandstone Member of the Ste. Genevieve Limestone, the Rosiclare ore in Ste. Genevieve Limestone beds beneath the Rosiclare Sandstone Member, and Bethel ore in limestones of the uppermost beds of the Downeys Bluff Member of the Renault Formation beneath the Bethel Sandstone Member (Baxter et al., 1992). The mantos range up to 3 km in length, 60 m in width, and 6 m in thickness; they occur next to major northeast-trending faults beneath cap-rock sandstones. The vein deposits are predominantly in the northeast-trending faults and mainly within the walls of the middle Mississippian strata. However, trace amounts of fluorite are found in faults cutting lower Pennsylvanian rocks. The vein deposits range up to about 250 m vertically and horizontally and up to 14 m in width.

The Paleozoic rocks have been intruded by Early Permian alkalic ultramafic dikes, sills, and igneous breccia plugs. About 60 dikes, a few sills, and nine plugs are known throughout the Illinois-Kentucky fluorspar district (Koenig, 1956; Nelson and Lumm, 1984). Dikes and sills are comprised of mica peridotite and lamprophyre. Most are <0.3 m thick, rarely >5 m, and nearly all dikes occupy steeply dipping (~75°) fractures that strike N 10°–45° W. Zartman et al. (1967) obtained K-Ar biotite ages of 252 ± 13

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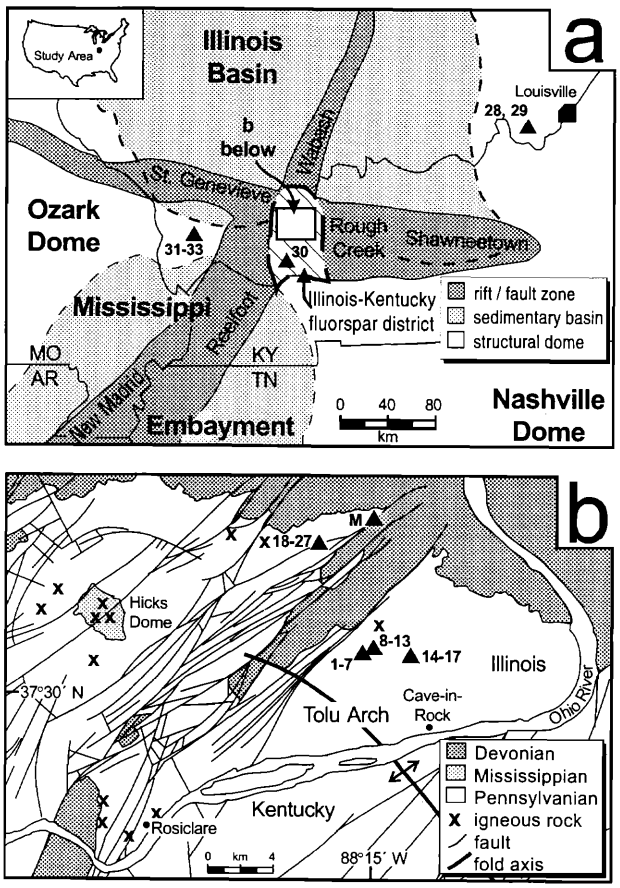


FIG. 1. Generalized geologic map of the study area (from Baxter et al., 1989) and site locations (triangles). Sites: 1–7, Lead adit of the Benzon mines in and beside a Rosiclare horizon orebody; 8–13, Green adit of the Austin-Cleveland mines in and beside a Rosiclare horizon orebody; 14–17, Plant 2 (Denny-Simpson quarry) of Hardin County Materials Company in Rosiclare horizon rocks; 18–27, Denton mine of the Ozark-Mahoning Company in and beside a Rosiclare horizon orebody; 28–29, roadcut, Indiana State Highway 135, 5.8 km south of Corydon, Indiana, in Ste. Genevieve Limestone; 30, open pit at the Hutson mine, 5.2 km southwest of Salem, Kentucky, in St. Louis Limestone wall rocks to vein; 31–33, roadcuts, Interstate Highway 57, 4.8 km north of Dongola, Illinois, exit in Rosiclare and sub-Rosiclare horizon equivalents in Ste. Genevieve Limestone; M, fluorite crystals, Annabel Lee mine of the Ozark-Mahoning Company.

and 269 ± 12 Ma and Rb-Sr biotite ages of 279 ± 27 and 260 ± 23 Ma for a dike and sill, respectively. Several diatreme occurrences are known including shatter, vent, and carbonatized igneous breccias around Hicks Dome (Fig. 1) and elsewhere in the district (Baxter et al., 1989). Barite, fluorite, sphalerite, and galena occur sparsely in some of these breccias. Zartman et al. (1967) determined K-Ar ages of 258 ± 13 Ma on biotite and 281 ± 14 Ma on hornblende xenocrysts from the Grants intrusive breccia. These were confirmed by Snee and Hayes (1992) who obtained concordant biotite and hornblende $^{40}Ar/^{39}Ar$ ages of 272 ± 1 Ma for the xenocrysts. In addition Reynolds et al. (1992) obtained Early Permian virtual paleomagnetic poles for the Grants diatreme and sill. Some dikes are cut by, and thus predate, the northeast-trending fluorspar-bearing block faults, thus setting a maximum stratigraphic age for the fluorspar mineralization (Grogan and Bradbury, 1968). After faulting, fluorite and other ore minerals were emplaced in veins cutting the dikes and in patches replacing dike material, notably at the Hutson mine (Koenig, 1956).

The Paleozoic rocks nearby in southern Illinois are unconformably overlain by about 1,300 m of Upper Cretaceous to Eocene sedimentary rocks, commencing with the conglomerates and gravels of the Tuscaloosa Formation (Pryor and Ross, 1962; Schwalb, 1982). These clastic rocks contain fluorite pebbles (A.V. Heyl, pers. commun. 1985, *in* Richardson et al., 1988). Further, nearly all the displacement on the northeast-trending faults predates deposition of the Tuscaloosa beds to set a minimum stratigraphic age for the fluorspar mineralization.

There have been three attempts at direct dating of the mineralization. In 1987, Vicki Harder (pers. commun., 1993) obtained fission track ages on fluorite from Cave-in-Rock mines of 135.7 ± 4.5 and 140.6 ± 4.7 Ma that straddle the Jurassic-Cretaceous boundary. Fluorite has an annealing temperature of about 108°C for one million years, so these ages probably indicate cooling of the fluorite after emplacement (Harder, 1986). The validity of using fluorite for fission track dating is controversial, having been supported, for example, by Koul and Chadderton (1988) and Gronlie et al. (1990) and questioned by Arne and Kelly (1992) and Talbot et al. (1992). The latter authors tried published methods of etching but were unable to produce tracks on Cave-in-Rock fluorite crystals. Ruiz et al. (1988) assumed that the fluorine and strontium were obtained from Early Permian igneous rocks so that their strontium isotope data indicate that fluorspar mineralization occurred in the Early Jurassic at about 200 Ma. Recently Chesley et al. (1993) have reported in an abstract a 14 point $^{147}Sm/^{144}Nd$ isochron of 272 ± 17 Ma for fluorites from Cave-in-Rock, indicating that mineralization is about coeval with emplacement of Hicks Dome and related Early Permian dikes and sills.

Structurally the fluorspar district is located at the juncture of the Mississippi embayment syncline to the southwest, the Nashville dome to the southeast, Illinois basin to the northeast and north, and Ozark uplift to the west (Fig. 1). The district is complexly faulted at the intersection of the northeast-trending New Madrid-Reelfoot-Wabash rift zone (Schwalb, 1982) with the high-angle reverse faults of the west-northwest-trending Ste. Genevieve-Rough Creek-Shawneetown zone. The Reelfoot rift is a graben

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structure that is filled in with Paleozoic sediments, thus forming a syncline. Similarly the Rough Creek fault bounds a graben, now to its south. Displacements of 30 to 150 m are typical on faults with fluorspar mineralization (Baxter et al., 1989). Seismic data indicate that the faults penetrate into the basement to provide access for mineralizing fluids (Goldhaber et al., 1992b; Potter et al., 1992). From structural analysis, Nelson (1992) considers the mineralization to occur in a dogleg-shaped Cambrian failed rift that was reactivated to generate the reverse faults in the late Paleozoic and again to generate the northeast-trending normal faults in the Mesozoic. Despite the numerous faults, dips of the strata in the Cave-in-Rock area of the district rarely exceed a few degrees including all sites in this study. The fluorspar mines lie astride the gentle Tolu arch (Fig. 1b). The axis of the arch is offset by the northeast-trending, ore-bearing faults, showing that the strata in the sampled area near Cave-in-Rock were dipping north-northeastward prior to being mineralized (Baxter et al., 1989).

The district's main ore mineral is fluorite. Sphalerite, galena, and barite are common and have been commercial accessory minerals at places. Calcite and dolomite are the main gangue minerals, commonly with minor quartz. Trace minerals include pyrite, marcasite, and chalcopyrite. The paragenetic sequence has been well described by Grogan and Bradbury (1968), Heyl (1983), Richardson and Pinckney (1984), Spry et al. (1990), and Hayes and Anderson (1992) who showed that the fluorite formed in several stages through the early to middle part of the sequence. Ore precipitation temperatures are well constrained in the Illinois-Kentucky fluorspar district. Richardson and Pinckney (1984) report homogenization temperatures in fluid inclusions ranging from 227° to 68°C in various stages of hydrothermal quartz and from 153° to 116°C in fluorite from mantos, with most between 153° and 140°C. Temperatures of about 150°C are two to three times greater than tenable given the maximum available thicknesses of about 2.5 km of overlying cover and an ordinary geothermal gradient (Grogan and Bradbury, 1968; Schwalb, 1982). This suggests that the mineralizing fluids came from deep in the basement or from an igneous source, and that the cooling was relatively rapid. Goldhaber et al. (1992), using metallic element concentrations determined by emission spectroscopy, Burruss et al. (1992), using regional and microscale zonation of the rare earth elements, and Taylor et al. (1992), using fluid inclusion temperatures of paragenetically equivalent fluorites, show gradients away from Hicks Dome across the Illinois-Kentucky fluorspar district that led them to suggest that the dome was a major source of heat and possibly of magmatic components for the mineralizing fluids. Conversely, Spry and Fuhrmann (1993) with a larger data base conclude that there is no regional inclusion temperature gradient and, therefore, question the role of the dome as a heat source or provider of ore-forming constituents.

From stable isotope analyses, Richardson et al. (1988) favor ore formation by mixing of fluids from two sources. They suggest that the early- to intermediate-stage ore fluids were very saline, having traveled through the basement leaching fluorine, lead, strontium, and possibly sulfur. Conversely, the late-stage ore fluids were relatively dilute or less saline, having evolved in the sedimentary section at shallow depths.

Sampling

Three oriented block samples were collected at each of 32 sites and four block samples at a thirty-third site (Fig. 1, Table 1). Eleven sites are from fluorspar ore—both vein and replacement zones—in adits of the Benzon and Austin-Cleveland mines and from underground in the Denton mine of the Cave-in-Rock area. Most of these specimens are massive fluorite, but some are disseminated ore. One site of ore-stage jasperoid was collected from the Denton mine. A further eight sites come from calcareous sandstones of the Rosiclare Sandstone Member either from cap rocks atop the ore zones or from nearby in the Hardin Materials Company's Plant 2 quarry. Finally 13 sites come from coeval middle Mississippian limestones. Of these 13, eight are from immediately adjacent to fluorspar ore and five are remote or tens of kilometers away from known significant mineralization. In addition three large unoriented fluorite crystals were obtained from the Annabel Lee mine.

Experimental Methods

From three to six specimens were drilled from each block sample giving a total of 324 specimens for analysis. After storage for two months in a magnetically shielded room, the natural remanent magnetizations (NRM) of the specimens were measured on an automated CTF cryogenic magnetometer with a noise level of $<4 \times 10^{-6}$ A/m. Initially two specimens from different blocks from each site were alternating field demagnetized in 12 steps to 120 mT using a Sapphire Instruments SI-4 alternating field demagnetizer. A third specimen from the third block from each site was thermally demagnetized in nine steps to 590°C using a Schonstedt TSD-1 demagnetizer. Most of the fluorspar specimens disintegrated by the 300° or 400°C step. Also alternating field step demagnetization proved much more efficacious at isolating the characteristic remanent magnetization (RM) components than did thermal step demagnetization. Therefore the remaining specimens were alternating field demagnetized, mostly in 15 or 20 mT steps above 20 mT. The characteristic RM directions for each speci-

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TABLE 1. Site Mean Remanence Data without Tilt Correction

Site	Lithology[1]	Strike, dip (°)	No. of specimens[2]			Demagnetizing range[3]		Remanence direction[4]				
			B	Su	Sr	mT	°C	Dec (°)	Inc (°)	k	α_{95} (°)	Cmp
1	Fgp	268, 18	2	7	0	10–120	300	358	35	302	3	B
2a	Fwp	267, 11	2	7	0	20–70	200–300	341	47	112	6	B
2b	Sfh	267, 11	1	2	0	15–120		8	67[5]	512	11	C
3	Fwb	283, 1	3	7	0	10–100	200–500	340	53	32	11	B
4	Fwb	283, 1	3	8	0	20–100		351	53	17	14	B
5	Sc	286, 4	3	7	1	20–100	300–560	7	49	11	19	B
6	Sc	284, 2	3	7	2	10–100		8	48	46	9	B
7	Lo	265, 5	3	9	0	10–80	400–580	354	50	34	9	B
8	Fpa	302, 8	3	8	1	20–80		351	41	16	14	B
9	Fpa	302, 8	3	8	1	20–80	200–300	4	42	26	11	B
10a	Fpgb	272, 10	2	7	0	20–80		351	22	40	10	B
10b	Lh	272, 10	1	4	0	20–120	450–560	221	70	42	14	
11	Sc	274, 10	2	8	4	20–80		330	28[5]	11	17	B
12	Lg	276, 7	3	11	0	15–100		341	35	11	14	B
13	Sc	260, 10	3	6	5	30–90	200–450	330	45[5]	17	17	B
14	Lg	280, 4	3	7	2	15–50	200–450	5	48	12	18	B
15	Sc	280, 4	3	8	2	30–50	200–400	345	37[5]	8	20	B
16	Sc	280, 4	3	10	0	20–120	400–560	348	−63[5]	45	7	
17	Sc	280, 4	3	7	2	20–50	400–450	339	41	11	19	B
18	Fgpt	290, 3	3	10	0	20–60	300–400	350	50	111	5	B
19	Jfpg	290, 3	3	10	0	10–50	300	348	51	31	9	B
20a	Fpg	290, 3	2	7	0	10–50	300–560	357	48	21	13	B
20b	Sc	290, 3	1	3	0	20–40		349	49	22	27	B
21	Lo	230, 5	3	9	1	15–100	300–450	338	47	52	7	B
22	Lo	280, 1	3	9	1	20–50	300–450	2	51	27	10	B
23	Lmg	280, 3	3	8	2	10–80	300–450	336	59	47	8	B
24	Fpg	240, 10	3	9	1	20–100	300–450	330	41	29	10	B
25	Fg	285, 3	3	11	0	15–100	300–450	354	40	86	5	B
26	Sc	295, 5	3	10	1	20–100	200–560	354	48	67	6	B
27	Lo	290, 1	3	11	0	20–80	300–530	354	53	117	4	B
28a	Lmg	100, 1	2	7	0	20–120	450–530	158	−5	10	20	A
28b	Lgh	100, 1	1	3	0	20–90		250	−40	262	8	
29	Lmg	100, 1	3	8	3	40–100	560–590	148	−7	13	16	A
30	Lmh	259, 4	4	12	1	20–120	450–530	348	45	32	8	B
31	Lmh	330, 7	3	6	4	0–120	300–500	5	70[5]	77	8	C
32	Lmh	62,2	3	9	1	0–120	300–500	13	76[5]	44	8	C
33a	Lmh	62,2	2	4	2	0–120	530–560	328	75[5]	33	13	C
33b	Lmh	62,2	1	4	0	20–60		197	68	118	8	

[1] Lithology: F = fluorspar, J = jasperoid, L = limestone, S = sandstone; a = barite, b = brown, c = calcareous, f = fluorite, g = gray, h = hematite, m = massive, o = oölitic, p = purple, w = white; see Figure 1 caption for stratigraphic levels sampled

[2] B = blocks represented, Su = specimen end points used, Sr = specimen data rejected

[3] Alternating field coercivities in millitesla (mT), unblocking temperatures in degrees Celsius (°C)

[4] Dec = declination, Inc = inclination, k = precision parameter, α_{95} = radius of cone of 95% confidence (Fisher, 1953) in degrees (°), Cmp = defined characteristic RM component

[5] Reversed characteristic RM directions used: site 11, two end points; site 13, one end point and one remagnetization circle (Bailey and Halls, 1984); site 15, two remagnetization circles; and site 16, ten remagnetization circles

men were defined using the principal component method of Kirschvink (1980), requiring the maximum angular deviation values to be <15°, with most <5°. Specimen characteristic RM directions were averaged to get the site mean directions (Fisher, 1953; Table 1). To characterize the magnetic mineralogy better, 12 representative specimens of the various ore and rock types and four mineral specimens were tested using saturation isothermal remanent magnetization methods. Using a Sapphire Instruments SI-6 pulse magnetizer, each specimen was magnetized in 12 steps to 900 mT and then alternating field demagnetized in five steps to 50 mT.

Results

The NRM intensities of the fluorspar ores are mostly in the 1 to 4×10^{-4} A/m range within a spectrum from 0.2 to 9×10^{-4} A/m. The jasperoid and calcareous sandstone specimens give NRM intensities that are mostly in the 2 to 5×10^{-4} A/m range;

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the limestones are in the 0.5 to 4 × 10^{-4} A/m range. The single crystals of fluorite give NRM intensities of 0.24, 0.32, and 1.24 × 10^{-4} A/m.

Alternating field demagnetization during the initial steps up to about 20 mT removes a normal remanence component that is directed northward at moderately steep inclinations in nearly all specimens. This component appears to be viscous remanent magnetization (VRM) from the present earth's magnetic field. This VRM also appears to be removed by about 300°C on thermal demagnetization.

For the fluorspar ore specimens alternating field demagnetization above about 15 or 20 mT isolates a well-defined characteristic RM component with an intermediate inclination downward to the north or northwest in virtually every specimen (Fig. 2). This is termed the "B" characteristic RM component. The *B* component is also isolated on thermal demagnetization by unblocking temperatures up to at least 530°C before disintegration of the specimens, suggesting that magnetite is the major remanence carrier in the ore. All six saturation isothermal RM acquisition curves for the massive fluorspar ore specimens are typical of single domain to pseudosingle domain magnetite (Fig. 3). Alternating field demagnetization of the saturation isothermal RM in these specimens similarly falls in the single domain to pseudosingle domain range for magnetite. The finding of magnetite as the only significant magnetic mineral is consistent with the thermal demagnetization results. The saturation isothermal RM acquisition curves from the three fluorite crystal specimens give much the same result, falling in the single domain to pseudosingle domain range. Note, however, that their alternating field demagnetization curves indicate more rapid decay and suggest only pseudosingle domain behavior. Also the intensities for the saturation isothermal RM of the crystal specimens range from 1.4 to 22.1 × 10^{-3} A/m, or about half an order of magnitude lower than the 6.2 to 70.2 × 10^{-3} A/m range of the fluorspar ore specimens. These differences are thought to reflect probable slower crystallization under more quiescent conditions for the crystals than for the massive ores, resulting in slightly coarser but fewer magnetite crystal inclusions in the fluorite single crystals relative to the ore.

The calcareous sandstone specimens from the cap rocks of the Rosiclare horizon also show the isolation of the *B* characteristic RM component, including domains with coercivities in excess of 120 mT, after ini-

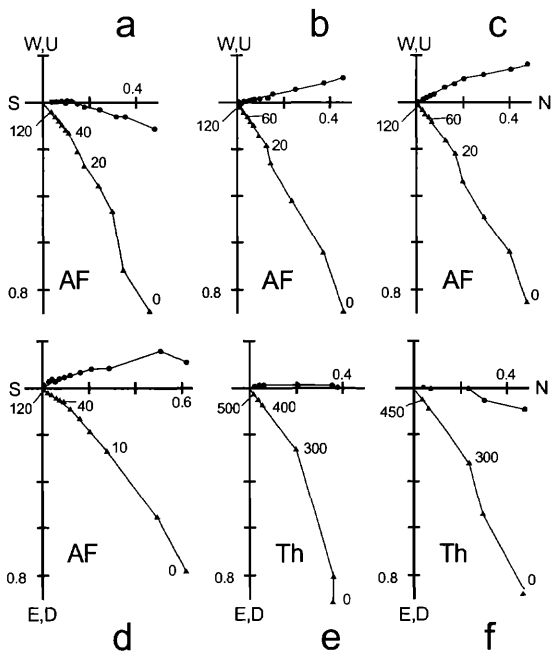


FIG. 2. Orthogonal demagnetization diagrams showing examples of fluorspar ore specimens carrying the *B* component on alternating field (AF) and thermal (Th) step demagnetizations from (a) site 1, gray fluorite; (b) site 18, purple fluorite; (c) site 24, purple fluorite; (d) site 25, gray fluorite; (e) site 3, white fluorite; and (f) site 25, gray fluorite. The axes are north (N), east (E), south (S), west (W), up (U), and down (D). Circles (triangles) denote projections onto the horizontal (vertical) plane. The axes show the J/J_0 ratio with the six J_0 values being 2.44, 6.95, 4.89, 9.77, 1.80, and 9.8 × 10^{-4} A/m, respectively. The alternating field steps are in millitesla and Th steps in degrees Celsius.

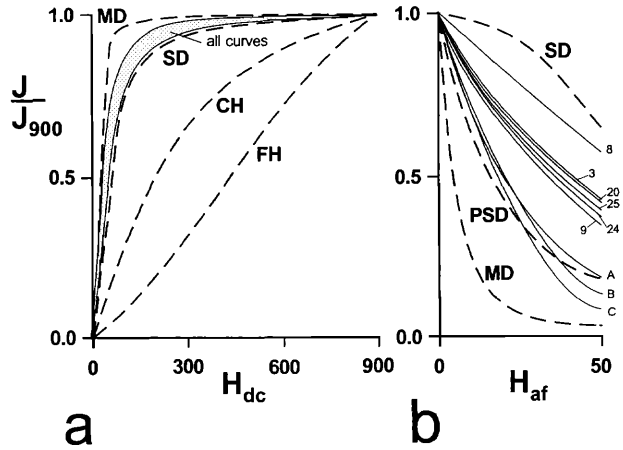


FIG. 3. Saturation isothermal remanent magnetization curves showing: (a) acquisition of saturation isothermal RM, and (b) demagnetization of saturation isothermal RM, for ore specimens from site 3, white and brown; site 8, purple; site 9, purple; site 20, pale purple; site 24, purple and gray; and site 25, gray fluorspar; and for single-crystal specimens of (A), clear; (B), pale yellow; and (C), purplish blue fluorite. J/J_{900} is the ratio of the measured remanence intensity to the saturation isothermal RM intensity at 900 mT, and H is the applied field in mT. The dashed line curves are typical for single domain (SD), pseudosingle domain (PSD), and multidomain (MD) magnetite and for fine-grained (FH) and coarse-grained (CH) hematite from Dunlop (1971, 1972, 1973, 1981). The saturation intensities, J_{900}, are 17.9, 13.4, 6.23, 38.7, 53.9, 70.2, 1.38, 1.92, and 22.1 × 10^{-3} A/m, respectively.

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tial VRM removal (Fig. 4). Thermal demagnetization results are similar to alternating field demagnetization results except that large unstable magnetization components are generated at temperatures over 530°C, probably as sulfide minerals are oxidized. Nevertheless, it is evident that magnetite and possibly hematite are present. One notable feature is that six specimens, two from each of sites 11, 13, and 15 (Table 1), gave reversed *B* directions as defined end points in three cases (Fig. 4c) and as remagnetization circles in three cases. The saturation isothermal RM acquisition curves for the sandstone specimens show an initial rapid increase that is indicative of magnetite and then a slow steady increase to 900 mT that is indicative of hematite (Fig. 5; sites 13, 15, and 17). There is some suggestion that in the highest temperatures reached on thermal demagnetization the remanence inclination is steepening before large unstable components are generated (e.g., Fig. 4d at 530°C, Fig. 4e at 560°C). This steepening could be caused by a modern remanence from weathering because the samples with evidence of hematite come from adit entrances and the Hardin quarry.

The near-ore limestones from the Cave-in-Rock fluorspar area and from the Hutson mine give results similar to the fluorspar ore and cap-rock sandstone specimens, giving the *B* characteristic RM component on both alternating field and thermal demagnetization after initial VRM removal (Fig. 6). This remanence is retained in grains with unblocking temperatures up to the 585°C Curie temperature of magnetite. This result is consistent with the saturation isothermal RM results where both the acquisition and alternating field demagnetization curves indicate only single domain to pseudosingle domain magnetite (Fig. 5; sites 7 and 27).

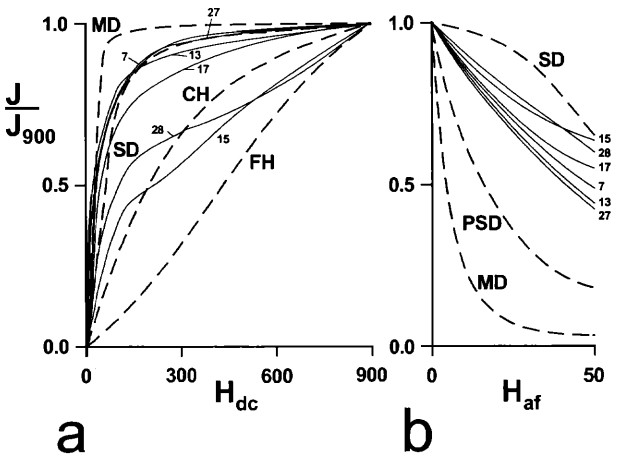


FIG. 5. Saturation isothermal RM curves for calcareous sandstone cap-rock specimens from sites 13, 15, and 17, and for limestone specimens from sites 17, 27, and 28 with J_{900} intensities of 13.0, 16.0, 19.4, 3.52, 2.58, and 1.56×10^{-2} A/m, respectively. Conventions as in Figure 3.

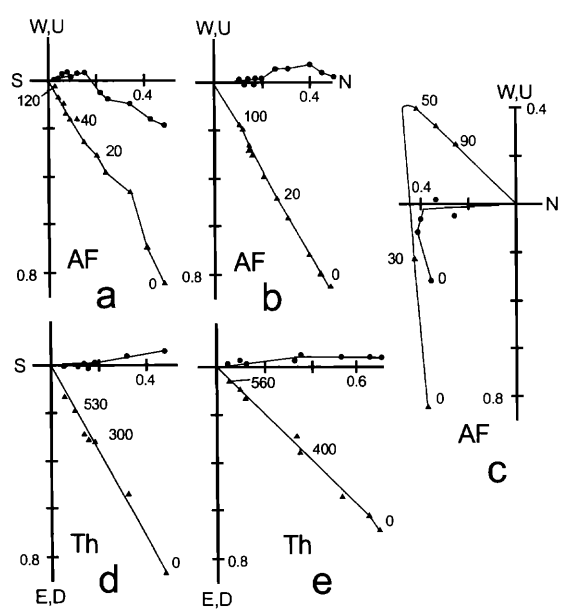


FIG. 4. Orthogonal demagnetization diagrams showing example specimens of calcareous sandstone cap rocks carrying the *B* component: (a) site 26, (b) site 6, (c) site 11, (d) site 26, and (e) site 5 with J_0 intensities of 2.69, 4.66, 2.76. 2.84, and 5.18×10^{-4} A/m, respectively. Conventions as in Figure 2.

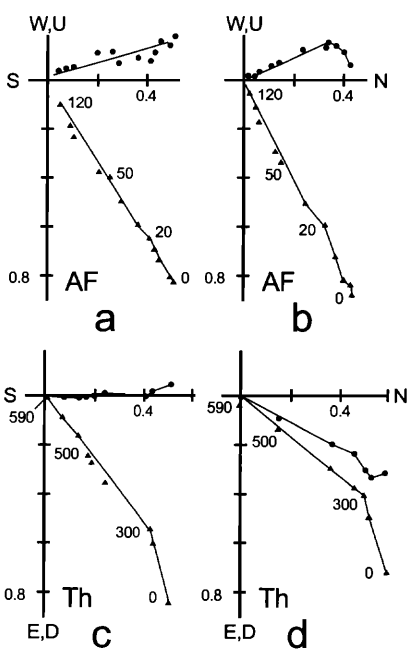


FIG. 6. Orthogonal demagnetization diagrams showing example specimens of near-ore limestone host rocks carrying the *B* component: (a) site 21, (b) site 23, (c) site 27, and (d) site 22. The J_0 intensities are 7.22, 4.39, 1.72, and 5.25×10^{-4} A/m, respectively. Conventions as in Figure 2.

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The limestone specimens from the control sites in road cuts remote from the district show radically different behaviors. The specimens from sites 28 and 29 from near Louisville, Kentucky, give a characteristic RM component directed about horizontally to the southeast that is termed the "A" component (Fig. 7a and b). This component is also retained in domains with unblocking temperatures that are typical of magnetite. However, occasional specimens from these sites also show evidence of hematite, probably from weathering (Fig. 5; site 28). In contrast, nearly all of the specimens from the roadcuts near Dongola, Illinois (Fig. 1, sites 31–33), give relatively steep downward directions to the north. This is termed the "C" characteristic RM component, and it has alternating field and thermal step demagnetization properties that are typical of hematite (Fig. 7c).

Finally, remanence and saturation isothermal RM measurements were made on a single-crystal galena specimen and on a combination of a plaster of paris blank specimen and plaster of paris plus about 10 percent by weight crushed galena. Unlike galena crystals from the Polaris mine (Symons and Sangster, 1992), the Cave-in-Rock galena does not retain a measurable NRM saturation remanence.

Discussion

C component

The interpretation of the *C* component is straightforward (Fig. 8). It records modern hematization from recent weathering at the Dongola roadcut (sites 31–33) and at the entrance to the Lead adit (site 2b). Note that the *C* pole position at 89.1° W, 69.3° N (dp = 12.9°, dm = 14.5°; Fisher, 1953) without stratigraphic tilt correction falls just south of the present earth's magnetic field direction or atop the north pole's path over the past few decades (Fig. 9, Table 2). Although only four sites are involved, use of the uncorrected site mean characteristic RM directions is supported by the fold test of Graham (1949). It shows with >95 percent confidence that the remanence is postfolding (McFadden and Jones, 1981).

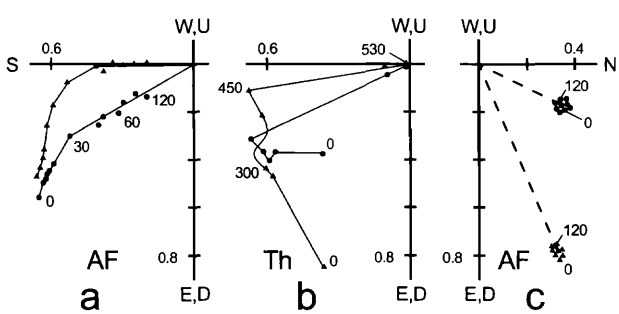


FIG. 7. Orthogonal demagnetization diagrams showing example specimens of remote limestone that is stratigraphically equivalent to ore that carries the *A* component: (a) and (b) site 28; and the *C* component: (c) site 32. The J_0 intensities are 0.46, 4.49, and 49.7 × 10^{-4} A/m, respectively. Conventions as in Figure 2.

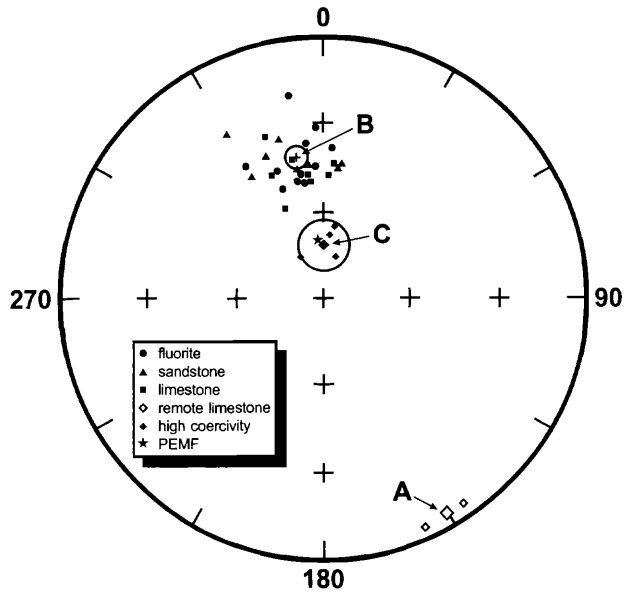


FIG. 8. Equal-area stereogram showing the site mean directions for fluorspar ore (circles), cap-rock sandstones (triangles), near-ore limestone host rocks (squares), and remote stratigraphically equivalent limestone (diamonds). Solid (open) symbols denote downward (upward) directional vectors. Larger symbols denote the *A* and *C* unit means; the cross denotes *B* unit mean direction. The *B* and *C* unit means are circumscribed by their cones of 95 percent confidence. The star is the present earth's magnetic field direction.

A component

The interpretation of the *A* characteristic RM component also appears straightforward. Found only at the two most remote limestone control sites (28a, 29) near Louisville, the *A* pole is located at 136° E, 47° N (dp = 11°, dm = 22°) after tilt correction (Fig. 1, Table 2). Tilt correction moves the pole by only one degree in any event. The *A* pole falls on the middle Pennsylvanian or Kiaman portion of the apparent polar wander path of Van der Voo (1991; Fig. 9). The *A* characteristic RM is typical of the reversed Kiaman remagnetization component that is found throughout much of the eastern and central midcontinent of the United States, including in other Mississippian and older rocks near Louisville (McCabe and Elmore, 1989). The remagnetization has been tied to uplift in the Appalachian and Ozark orogenic belts during the Alleghanian orogeny that caused the regional migration of brines from adjacent basins into the surrounding platforms (Garven and Freeze, 1984a,b; Leach and Rowan, 1986; Oliver 1986). The brines are thought to be responsible for the formation of Mississippi Valley-type ore deposits of lead, zinc, and barite that are generally akin to the Illinois-Kentucky

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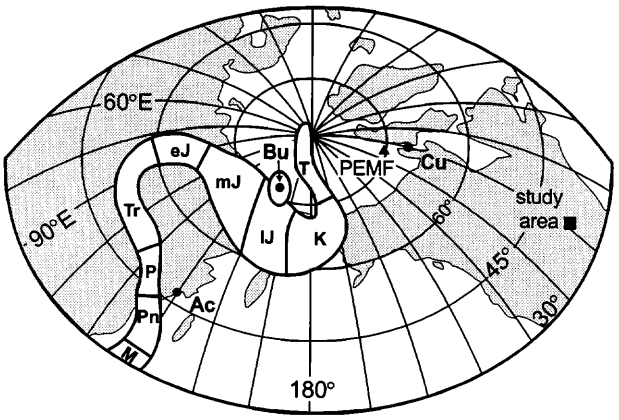


FIG. 9. Apparent polar wander path for the North American craton with pole positions. The path for the Mississippian (M), Pennsylvanian (Pn), Permian (P), Triassic (Tr), and Early Jurassic (eJ) is from Van der Voo (1990), that for the Middle Jurassic (mJ) and Late Jurassic (lJ) is simplified from Van der Voo (1992), and that for the Cretaceous (K) and Tertiary (T) is from Irving and Irving (1982). The pole for the *A* characteristic RM is tilt corrected (Ac), for *B* is uncorrected (Bu), and for *C* is uncorrected (Cu) (Table 2). The *B* pole is circumscribed by its oval of 95 percent confidence, and the *C* pole from modern weathering is next to the present earth's magnetic field (PEMF).

fluorspar deposits. Mississippi Valley-type deposits with a Kiaman remanence include the southeast Missouri, central Missouri and northern Arkansas Tri-State districts in the Ozark dome to the southwest and the Eastern Tennessee district to the southeast (Wisniowiecki et al., 1983; Bachtadse et al., 1987; Pan et al., 1990; Symons and Sangster, 1991).

B component

The *B* component carried by the fluorspar ore and nearby host rocks in the Illinois-Kentucky district requires closer scrutiny. Using the uncorrected site mean directions, it is evident that the mean directions for the ore sites, cap-rock sandstone sites, and near-ore host limestone sites fall within each other's cone of 95 percent confidence (Table 2). The same is true for the corrected site mean directions. This indicates and the test of McFadden and Lowes (1981) confirms that the three *B* populations are statistically indistinguishable with ⩾95 percent confidence and can be combined to give an overall *B* unit mean direction (Table 2).

When the fold test is applied to the *B* component site mean directions, the change in dispersion is not statistically significant at the 95 percent confidence level (McFadden and Jones, 1981), but the precision parameter, k, does decrease from 69 to 53 on unfolding. Thus the bias is to favor using the uncorrected data. The corrected *B* mean directions are statistically different from the uncorrected ones because all but one of the sites are on the northeastern side of the Tolu arch with a dip of about 4° NNE on average throughout the Cave-in-Rock area. Relying on the geologic evidence which shows that the Tolu arch predates mineralization, the uncorrected site mean directions are deemed to provide the best estimate for the characteristic RM of the fluorspar mineralization. They give a unit pole position of 152.4° E, 78.0° N (dp = 3.0°, dm = 4.4°).

The location of the Jurassic apparent polar wander path for North America has been the subject of considerable recent debate (Hagstrum, 1993). Irving and Irving (1982) used a 30-m.y. moving average of all available poles to define a Jurassic apparent polar wander path which tracks about along the middle of the path shown in Figure 9. Gordon et al. (1984) and May and Butler (1986) used a paleomagnetic Euler

TABLE 2. Unit Mean Characteristic Remanent Magnetization Directions and Pole Positions

	Mean remanence direction						Pole position			
Unit; sites	*n*	Dec (°)	Inc (°)	k	α_{95} (°)	Comp	Long (°E)	Lat (°N)	dp (°)	dm (°)
Fluorspar ore; 1, 2a, 3, 4, 8,	12	348.2	50.1	86	4.7	Bu	150.7	78.2	4.2	6.3
10a, 18, 19, 20a, 24, 25	12	349.8	44.0	54	6.0	Bc	130.8	75.4	4.7	7.5
Caprock Sandstone; 5, 6, 11, 13,	8	345.2	48.2	42	8.7	Bu	152.1	75.1	7.4	11.3
17, 20b, 26	8	346.7	44.0	39	9.0	Bc	139.4	73.7	7.0	11.3
Near-ore Limestone; 7, 12, 14, 21,	8	350.1	52.5	90	5.9	Bu	156.1	80.8	5.6	8.1
22, 23, 27, 30	8	349.7	48.9	67	6.8	Bc	142.9	78.5	5.9	9.0
B component sites; all above	28	347.8	50.3	69	3.3	Bu	152.4	78.0	3.0	4.4
	28	348.9	45.4	53	3.8	Bc	136.3	75.8	3.1	4.8
A component; 28a, 29	2	153	−5	127		Au	135	47	11	22
	2	153	−6	128		Ac	136	47	11	22
C component; 2b, 31, 32, 33a	4	0.1	72.7	127	8.2	Cu	270.9	69.3	12.9	14.5
	4	5.3	70.1	43	14.2	Cc	283.4	72.9	21.1	24.6

Notes: *n* = number of sites; Dec, Inc, k, and α_{95} as in Table 1; Comp = component before (u) and after (c) tilt correction; Long = longitude, degrees east; Lat = latitude, degrees north; dp, dm, semi-axes of the oval of 95 percent confidence about the pole (Fisher, 1953)

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pole model for North American motion and heavy reliance on poles from units in the western United States to define a lower latitude path that follows closer to the southern side of the path in Figure 9. Van Fossen and Kent (1990), relying more heavily on poles for units in the eastern United States, have argued for a higher latitude path that follows closer to the northern side of the Figure 9 path. Van der Voo (1992) used all available poles of reasonable quality from North America plus rotated poles from other continents with allowance for post-Jurassic tectonic motion. His irregular path was smoothed to give the "swallowing-snake"-shaped path shown in Figure 9. Relative to this path the uncorrected *B* pole defines a Late Jurassic age of about 155 ± 10 Ma. Acceptance of a more northerly path (e.g., Van Fossen and Kent, 1990) would increase the age to about 160 Ma and a more southerly path (e.g., Gordon et al., 1984) would decrease the age to about 150 Ma, still within the Late Jurassic. The ±10 m.y. error is estimated from the 95 percent error oval about the pole.

Geologic evidence requires that the fluorspar mineralization in the Illinois-Kentucky district be younger than the Early Permian ultramafic dikes which have been altered and cut by fluorite mineralization. Geochemical evidence of regional zoning from the metallic and trace elements, from the several stages of fluorite in the paragenetic sequence and from the fluid inclusion temperatures in the ore minerals, favors a close genetic relationship of mineralization to intrusion at Hicks Dome in the Early Permian (Burruss et al., 1992; Goldhaber et al., 1992a; Taylor et al., 1992), although this evidence is disputed (Spry and Fuhrmann, 1993). Conversely, structural evidence favors a Mesozoic origin (Nelson, 1992). Sufficient time had to elapse for the regional stress field to change from the northeast-southwest extension pattern that predominantly controlled dike emplacement at 272 Ma to the northwest-southeast extension pattern that predominantly formed the northeast-trending normal faults, as well as additional time for mineralizing these faults. Some district faults have been subjected to reactivation. Most of the motion predates deposition of the Late Cretaceous conglomerates with fluorite pebbles; this sets a minimum age for the fluorspar ore. Fluorite is a brittle mineral, yet the fluorite in the Cave-in-Rock area shows little evidence of being crushed by fault reactivation, implying that it was emplaced shortly after Pascola arch deformation but before the Late Cretaceous (Baxter et al., 1989).

Direct radiometric dating by $^{147}Sm/^{144}Nd$ of the fluorites gives an age of 252 ± 17 Ma, coincident with the 272 ± 1 Ma age by $^{40}Ar/^{39}Ar$ for the Grants breccia and similar K/Ar and Rb/Sr ages from breccia plugs, dikes, and sills in the area (Zartman et al., 1967; Snee and Hayes, 1992; Chesley et al., 1993). $^{147}Sm/^{144}Nd$ appears to have given the correct age for granite-hosted fluorite veins in Cornwall; however, much of the theory is uncertain, including processes of rare earth element fractionation in fluorites (Chesley et al., 1991). Strontium isotopes provide an age of about 200 Ma (Ruiz et al., 1988); however, this result is imprecise with an error that is probably a few tens of millions of years. Finally, the fluorite fission track ages of V. Harder (pers. commun., 1993) are quite precise, indicate an age of slightly over ~138 ± 5 Ma, but they cannot be duplicated according to Talbot et al. (1992). Obviously the paleomagnetic age provides a good fit to the fission track and structural evidence, a dubious fit to the strontium isotope age, and a conflict with the $^{147}Sm/^{144}Nd$ age.

There are several consequences of the proposed Late Jurassic age for the Illinois-Kentucky fluorspar district that merit comment. One is that this remagnetization and mineralization event, hereafter termed the "hydrothermal Rosiclare event," possibly extends over the entire district. Second, the Rosiclare event is an island surrounded by rocks that record the Pennsylvanian-Permian Kiaman event so that it is probable that the Rosiclare event remagnetized rocks that were themselves remagnetized earlier in the Kiaman event. Third, there is no obvious Late Jurassic orogenic event to drive deep-seated basinal fluids into the area to generate the Rosiclare magnetization. There are no known Jurassic intrusions of consequence within about 100 km to provide an igneous source. However, it is clear from the beveling of Paleozoic sediments by erosion and from the initial deposition of coarse clastics in the Mississippi embayment at the start of Late Cretaceous time that substantial uplift in middle Mesozoic time must have occurred. The most obvious candidate is the Pascola arch, centered on the Reelfoot rift about 150 km southwest of southern Illinois and extending into Illinois. From stratigraphic data Kolata and Nelson (1991) estimate uplift at between 2.4 and 4.6 km. Similarly, Leach et al. (1994) set uplift at between 2.1 and 3.7 km using conodont alteration index values. Arguing that some time must be left for erosion of the uplift prior to overlying Late Cretaceous sedimentation, Kolata and Nelson (1991, fig. 1) suggest an end to uplift in mid-Jurassic time. This uplift must be assumed to have been in response to large intraplate tectonic forces and to have caused a major period of reactivation on the fluorspar district faults. The speculation is, therefore, that the Rosiclare event coincides with the Mesozoic uplift event with concomitant fault reactivation and northeasterly fluid flow in the Reelfoot-Wabash rift zone. Some support comes from Bethke et al. (1991) who found that oil in late Paleozoic reservoirs in central Illinois correlated geochemically and isotopically with bitumen from Devonian-Mississippian source rocks in

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southern Illinois. They argued for regional northward flow in the Mesozoic off the uplifted Pascola arch, reaching a maximum in the Early Cretaceous. Richardson et al. (1988), Spry et al. (1990), and Spry and Fuhrmannn (1993) found that mixing of two hydrothermal fluids was necessary for fluorspar genesis. An obvious speculation is that hot brines moving northeasterly from deep in the Reelfoot rift on Pascola arch uplift (Goldhaber et al., 1992b; Leach et al., 1994) may have met and mixed with lower temperature brines at the juncture with the crosscutting Ste. Genevieve-Rough Creek fault zone. Fourth, it is not realistic to hypothesize that the fluorspar deposits were emplaced at 272 Ma only to have their characteristic RM reset by heating in the Late Jurassic. The rank of the Pennsylvanian coal beds in the area indicates only 1,300 m of burial (Schwalb, 1982) and the dominant fluid inclusion temperatures of 140° to 153°C also limit postemplacement heating to less than about 100°C. Such temperatures are far too low to have reset a single domain to pseudosingle domain remanence in magnetite with unblocking temperatures approaching the 585°C Curie temperature of magnetite (Pullaiah et al., 1975; Walton, 1980). Fifth, these results support models for ore genesis that invoke subsurface brine flows (e.g., Richardson et al., 1988) and oppose models that invoke fluids associated with the 272 Ma Hicks Dome intrusive events (e.g., Taylor et al., 1992). In this context Leach and Sangster (1994) associate the genesis of the southern Illinois and other midcontinent fluorite deposits with extensional basin-continental rift tectonics, noting that some of the other deposits are not associated with any known igneous intrusions. Sixth, this study may add another high-latitude Jurassic pole for eastern North America (Van Fossen and Kent, 1990), although it should not be used for apparent polar wander path construction until confirmed by another dating method.

Conclusions

The results of this paleomagnetic study of fluorspar ores and their host rocks in the Illinois-Kentucky district lead to the following conclusions:

1. Mississippian limestones from some 200 km to the northeast near Louisville, which are stratigraphically equivalent to the host rocks, retain an *A* characteristic RM component. It resides in single domain to pseudosingle domain magnetite and has a reversed polarization. *A* gives a mid-Pennsylvanian pole position that is typical of Kiaman remagnetization of older Paleozoic limestones throughout much of the U. S. midcontinent.

2. During the Late Jurassic, the district's limestone host rocks and sandstone cap rocks, which probably carried a Kiaman remanence, were remagnetized to carry the *B* characteristic RM. Coincidentally the fluorspar ore was emplaced with the *B* characteristic RM. *B* resides only in single domain to pseudosingle domain magnetite, and the *B* pole is located at 152.4° E, 78.0° N (dp = 3.0°, dm = 4.4°). This Late Jurassic Rosiclare hydrothermal event suggests, along with pre-Late Cretaceous erosion beveling, a period of major uplift in the area during the Late Jurassic. Correlation of uplift with the elevation of the Pascola arch appears probable. Uplift was accompanied by significant reactivation and displacement of the faults of the Reelfoot-Wabash rift system as well as hydrothermal fluid migration and ore deposition in them.

3. The third or *C* characteristic RM component was found at the surface only, and it records hematization from recent weathering in the host rocks.

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